The parameter restrictions in Table 1

A source-table calculation for Leemans–Vandenschrick

The first case of Section 5.3 of the source has odd characteristic and parameters $a, a' \notin \{0, 1, -1\}$. Its four involutions preserve a nonsingular conic. When both maximal rank-three subgroups are S_4 or A_5, Table 1 records the possible characteristics for the three adjacent orders. We recover every numerical cell below. This is a necessary parameter calculation; it neither proves the intersection property nor replaces the separately acquired classification of the resulting groups.

Write X for the characteristic-polynomial variable. Direct multiplication of the four displayed matrices on physical page 16 gives

$$(X-1)(X^2-2aX+1), \quad (X-1)(X^2+CX+1), \quad (X-1)(X^2-2a'X+1),$$
$$C = -aa' + a + a' + 1.$$

Thus the source's b in the last characteristic polynomial on physical page 17 must be a'. For end order three, $a = -1/2$; for end order five,

$$a = \frac{-1+t}{4} \quad \text{or} \quad \frac{-1-t}{4}, \qquad t^2 = 5.$$

The first assertion follows from the two primitive cube roots; the second from $2a = \zeta + \zeta^{-1}$ for a primitive fifth root. In the present nondegenerate parameter case characteristic five is excluded from an order-five end, since it would give $a = 1$.

The adjacent order in the middle is three, four or five precisely when, respectively,

$$C = 1, \qquad C = 0, \qquad C^2 - C - 1 = 0. \tag{1}$$

For the semisimple cases this follows from the reciprocal eigenvalue pair. In characteristic three or five the repeated eigenvalue one requires a qualification: the two central involutions are distinct, so their product is not the identity. Its nontrivial unipotent part in three dimensions has order three or five, respectively. Their scalar centres cannot shorten the order, because the matrices have eigenvalue one. Thus (1) remains valid in the relevant characteristics.

An order-four middle cannot be adjacent to an order-five end in S_4 or A_5: the former contains no element of order five and the latter none of order four. These are the dashes in the source table; they are not omitted equations. An order-four middle with two order-three ends would require $C = -1/4 = 0$, which is impossible.

For the other cells, substitute the three possible values of each end parameter. Symmetry in a, a' and replacement $t \mapsto -t$ reduce the nine source rows to the following four forms. The last two columns are possible characteristics; the actual equation in (1) must still hold for the chosen root t in the ambient field.

(a, a')	C	middle 3	middle 5
$(-1/2, -1/2)$	$-1/4$	5	11
$(-1/2, (-1 \pm t)/4)$	$(1 \pm 3t)/8$	$\varnothing$	59
$((-1 \pm t)/4, (-1 \pm t)/4)$	$(1 \pm 5t)/8$	19	$3, 11$
$((-1+t)/4, (-1-t)/4)$	$3/4$	$\varnothing$	19

For clarity, these restrictions follow by integer elimination rather than a search over fields. Reduce the left sides of (1) modulo t^2-5 and clear powers of two. Their integer numerators u_3, u_5, in the same four rows and using the plus choice of sign, are

	u_3	$\mathrm{Res}_t(t^2-5,u_3)$	u_5	$\mathrm{Res}_t(t^2-5,u_5)$
1	-5	25	-11	121
2	$3t-7$	4	$-9t-13$	-236
3	$5t-7$	-76	$27-15t$	-396
4	-1	1	-19	361

(2)

A common root in a field of odd characteristic forces that prime to divide the corresponding integer resultant. Factoring (2) gives exactly the displayed characteristic lists. Conversely, in each listed characteristic the two polynomials have a common root in a finite extension, and it does not force either end parameter into $\{0,1,-1\}$. The supplied exact polynomial calculation checks those small gcd assertions and retains all nine signed rows. The ambient field must contain this common root; one may not assert that both choices of $\sqrt{5}$ work merely from the prime.

This recovers the parameter restrictions in the source table and corrects the last displayed characteristic polynomial. Its title must not be read as a certificate that every listed parameter choice gives a rank-four C-group. The actual generation and intersection conditions are supplied by the separately reviewed conic classification. The calculation adds no whole-original, conjecture, or measurement credit.